Module7_Lab2

October 17, 2026

[1]:
```
import pandas as pd

df = pd.read_csv('ratings.tsv', sep='\t')
print(df.head())
print(df.info())
```

```
   restaurant  food  ambience  service  rating
0           1    85        82       89      78
1           2    80        90       80      85
2           3    83        86       83      85
3           4    70        96       75      72
4           5    68        80       78      75
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   restaurant  15 non-null     int64
 1   food        15 non-null     int64
 2   ambience    15 non-null     int64
 3   service     15 non-null     int64
 4   rating      15 non-null     int64
dtypes: int64(5)
memory usage: 732.0 bytes
None
```

[2]:
```
X = df[['food', 'ambience', 'service']]
y = df['rating']

model = LinearRegression()
model.fit(X, y)

coefficients = pd.Series(model.coef_, index=X.columns)
print("Coefficients:")
print(coefficients)

ranked_coefs = coefficients.sort_values(ascending=False)
print("\nRanked coefficients (highest to lowest slope):")
```

```
print(ranked_coefs)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[2], line 4
      1 X = df[['food', 'ambience', 'service']]
      2 y = df['rating']
      3
----> 4 model = LinearRegression()
      5 model.fit(X, y)
      6
      7 coefficients = pd.Series(model.coef_, index=X.columns)

NameError: name 'LinearRegression' is not defined
```

[3]:
```
import pandas as pd
from sklearn.linear_model import LinearRegression

X = df[['food', 'ambience', 'service']]
y = df['rating']

model = LinearRegression()
model.fit(X, y)

coefficients = pd.Series(model.coef_, index=X.columns)
print("Coefficients:")
print(coefficients)

ranked_coefs = coefficients.sort_values(ascending=False)
print("\nRanked coefficients (highest to lowest slope):")
print(ranked_coefs)
```

```
Coefficients:
food        0.733776
ambience    0.133761
service     0.323981
dtype: float64

Ranked coefficients (highest to lowest slope):
food        0.733776
service     0.323981
ambience    0.133761
dtype: float64
```

For a struggling restaurant, Food should be paid attention to and fixed before anything else according to the ranked coefficient. The highest ROI is Food at 0.733776, meaning for every single-unit increase in the food rating, the overall restaurant rating increases as well.